

ECONOMIC ORDER QUANTITY (EOQ) & SAFETY STOCK PARAMETERS

Date Prepared: October 26, 2023 | Prepared By: Supply Chain Analytics Team

1. Purpose

This document provides the foundational data, variables, and calculations required to determine the optimal order quantities (EOQ) and necessary safety stock levels for our key inventory items. By accurately calculating these figures, the company aims to minimize total inventory holding costs and ordering costs while preventing stockouts.

2. Calculation Methodologies

- Economic Order Quantity (EOQ): The formula used is $EOQ = \sqrt{(2 * D * S) / H}$
D = Annual Demand (units)
S = Ordering Cost per order (\$)
H = Holding Cost per unit per year (\$)
- Safety Stock: The formula used is $Safety\ Stock = (Max\ Daily\ Demand * Max\ Lead\ Time) - (Average\ Daily\ Demand * Average\ Lead\ Time)$

3. Baseline Assumptions

- Working Days per Year: 250 days
- Standard Ordering Cost (S): \$50.00 per order (includes administrative time, processing, and receiving labor).
- Holding Cost Percentage: 20% of the unit cost per year (includes warehousing, insurance, obsolescence, and opportunity cost of capital).

4. Parameter Breakdown by SKU

Parameter / Metric	GIZ-101 (Gizmo Model A)	WID-003 (Cardboard Packaged Widget)
Current BOH	85 units (Critical)	155 units
Unit Cost	\$15.00	\$3.00
Holding Cost (H)	20% of \$15.00 = \$3.00/unit/year	20% of \$3.00 = \$0.60/unit/year
Annual Demand (D)	2,500 units	5,000 units
Average Daily Demand	10 units	20 units
Max Daily Demand	18 units	30 units
Average Lead Time	14 days	7 days
Max Lead Time	21 days	10 days
EOQ Calculation	$[(2 * 2500 * 50) / 3] = [250000 / 3] = 83333.33$ approx. 289 units	$[(2 * 5000 * 50) / 0.60] = [500000 / 0.60] = 833333.33$ approx. 913 units
Safety Stock Calculation	$(18 * 21) - (10 * 14) = 378 - 140 = 238$ units	$(30 * 10) - (20 * 7) = 300 - 140 = 160$ units
Reorder Point (ROP)	$140 + 238 = 378$ units Immediate Order Required (BOH is 85)	$140 + 160 = 300$ units Immediate Order Required (BOH is 155)

5. Conclusion

Based on the calculations above, draft purchase orders must be generated immediately for SKUs GIZ-101 and WID-003 to replenish stock back to optimal levels and prevent imminent stockouts. The order quantities on the draft PO should reflect the calculated EOQ values.